

Pathologic Studies of Fatal Encephalomyelitis in Children Caused by Enterovirus 71

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Key Words: Enterovirus 71; Encephalomyelitis; Viral transmission; Immunohistochemistry

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ABSTRACT

Objectives: Enterovirus 71 (EV71) is the major pathogen of hand, foot, and mouth disease and can cause death; however, its pathogenesis remains elusive.

Methods: We performed a detailed systematic histopathologic examination and molecular studies on six autopsy cases of EV71 infection using H&E, immunohistochemistry, double immunofluorescence staining, and nested reverse transcription polymerase chain reaction.

Results: Characteristic features of acute encephalomyelitis were observed. Viral antigens were mainly detected in neuronal cytoplasm and processes in the different brainstem nuclei and spinal cord, including the anterior and posterior horn cells. Viral antigens were also positive in the nerve roots of spinal cord and autonomic ganglia of intestines.

Conclusions: Our study revealed direct pathologic evidence supporting viral entry into the central nervous system (CNS) through peripheral nerves. In addition to the major motor pathway, EV71 can also enter the CNS via peripheral sensory and autonomic pathways in retrograde axonal transport.

Upon completion of this activity you will be able to:

- describe the pathologic features of fatal enterovirus 71 (EV71) infection.
- discuss the model of EV71 spread into the central nervous system.
- list cellular receptors thought to be involved in EV71 infection of cells.

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Enterovirus type 71 (EV71) is a single-stranded positive RNA virus belonging to the human enterovirus species A within the family Picornaviridae.¹ EV71 mostly affects children and causes hand, foot, and mouth disease (HFMD). Some patients with HFMD caused by EV71 infection can have severe neurologic complications, including aseptic meningitis, acute flaccid paralysis, and fatal encephalomyelitis, which is associated with systemic features, such as severe pulmonary edema and shock.²⁻⁶ Since it was initially isolated in California in 1969,⁷ EV71 has become the major pathogen of HFMD,⁸ causing many large HFMD outbreaks with significant mortalities throughout the world, especially in the Asia-Pacific region.^{2-6,9-11}

The pathogenesis of EV71 infection, especially the model of viral spread into the central nervous system (CNS), is still poorly defined. Autopsy studies have revealed some features of acute encephalomyelitis, manifesting as inflammation and viral antigens mainly in the brainstem and spinal

cord.^{2,5,6,12-15} The thalamus, hypothalamus, basal ganglia, dentate nucleus, and motor cortex were occasionally involved.^{3,16-18} Possible retrograde spread of EV71 through peripheral nerves into the CNS was proposed.^{16,19,20} In vivo experiments showed that EV71 can directly infect the brainstem via cranial nerves in mice,²¹ but no such direct evidence in human cases has been reported.

Viral receptors are the primary determinants of tissue tropism and play a crucial role in the pathogenicity of viruses.²² Several receptors for EV71 have been identified, including human scavenger receptor class B, member 2 (SCARB2)²³; human P-selectin glycoprotein ligand 1²⁴; annexin II²⁵; and sialylated glycans.²⁶ SCARB2 localizes mainly to lysosomes and shows the highest efficiency for viral infection.²⁷ In vivo experiments demonstrated that the exogenous expression of human SCARB2 in mice confers susceptibility to EV71 infection.²⁷ Few reports have studied the distribution of SCARB2 in tissues of fatal EV71 cases¹⁵ and uninfected patients.²⁸ To our knowledge, no published information has examined the systematic distribution of SCARB2 and the colocalization of SCARB2 with viral antigens in fatal EV71 cases, which is critical for understanding the EV71 pathogenesis.

In recent years, several large HFMD outbreaks associated with many fatal cases have been reported in Mainland China.^{9,10,12} The purpose of this study is to investigate the possible pathways of EV71 spreading into the CNS, the cellular targets of EV71 infection, the systematic inflammatory response, and the role of SCARB2 in the infection in a series of autopsy cases performed during these outbreaks using H&E, immunohistochemistry (IHC), double immunofluorescence staining, and nested reverse transcription polymerase chain reaction (RT-PCR).

Materials and Methods

Patient Information and Preparation of Human Tissue Samples

Samples from a total of six patients (Table 1) who died of EV71 infection, collected at the Department of Forensic Medicine, Tongji Medical College of Huazhong University of Science and Technology from 2010 to 2013, were used in this study. Samples from a 3-year-old boy who died of acute cardiac trauma without EV71 infection were used as a negative control. Fresh tissues, body fluid, and different swabs from each case were obtained for viral identification and isolation. All tissues were fixed in 10% buffered formalin for a week. The cerebral hemispheres were cut into 1-cm coronal slices after separation from the midbrain. The brainstem was cut at a 2-mm thickness horizontally in series, and the cerebellum was cut perpendicularly to the long axis. The spinal

cord with attached peripheral nerve roots was sampled. Tissue blocks from the non-CNS organs were sampled routinely (2 × 2 × 0.2 cm). All tissues blocks were dehydrated and embedded in paraffin. Serial sections of 5 μm thickness were used for H&E, IHC, and double immunofluorescence staining. In the negative control case, the tissue specimens were sampled and tested in the same way as the confirmed six cases. However, viral identification and isolation were not performed. All experimental protocols involving human samples were approved by the patients' relatives and the Human Ethical Commission at Tongji Medical College of Huazhong University of Science and Technology.

Histopathology and Immunohistochemistry

H&E staining was processed routinely. Primary antibodies used for IHC are listed in Supplemental Table 1 (all supplemental materials can be found at *American Journal of Clinical Pathology* online). After deparaffinization, all sections were dipped into 3% hydrogen peroxide solution for 10 minutes to quench endogenous peroxidase. Microwave-induced epitope retrieval was performed in citrate buffer (pH 6.0) at 100 °C for 30 minutes. After incubation for 30 minutes with 5% bovine serum albumin to block nonspecific binding, the sections were incubated with primary antibody in phosphate-buffered saline at 4 °C overnight. Then sections were incubated for 30 minutes at 37 °C with biotinylated second antibody and horseradish peroxidase-conjugated streptavidin (DAKO, Glostrup, Denmark), respectively. Staining was visualized with the 3,3'-diaminobenzidine chromogen. Finally, sections were counterstained with hematoxylin. Negative control sections were incubated with non-immune serum of the species from which the primary antibodies were derived.

Double Immunofluorescence Staining

Briefly, after deparaffinization, antigen retrieval, and blockage as mentioned above, the sections were incubated with both primary antibodies for mouse anti-EV71 and rabbit anti-SCARB2 overnight at 4 °C, followed by a mixture of Alexa 488-conjugated anti-mouse immunoglobulin G (IgG) and Alexa 555-conjugated anti-rabbit IgG (both 1:400; Abcam, Carlsbad, CA) for 1 hour at 37 °C. Control sections were incubated with non-immune serum to replace the primary antibodies. Sections were counterstained with 4',6-diamidino-2-phenylindole (DAPI) (1:50; Sigma-Aldrich, St Louis, MO). Digital images were obtained using confocal microscopy (A1 plus; Nikon, Tokyo, Japan).

Nested RT-PCR Detection of EV71 RNA and Virus Isolation

Nested RT-PCR detection of EV71 RNA was performed in all fresh tissues, body fluids, and swabs. Virus RNA was

Table 1
Patient Data^a

Case No.	Year	Age, mo/Sex	Origin	Clinical Presentation	Positive IHC of Viral Antigens
1	2010	24/M	Xiangfan Hubei	Fever, rash on four limbs and crissum, mouth ulcers, hydrostomia, and poor feeding for 2 d; 17 h after admission, he developed cyanosis, cold extremities, and tachypnea and then deteriorated rapidly to death	Brainstem, spinal cord, dentate nucleus, hypothalamus, basal ganglia, motor cortex, glandular epithelium, and autonomic ganglia of intestines
2	2011	10/F	Xiangfan Hubei	Lethargy, fever, and convulsion for 4 d associated with rash on four limbs and trunk, lower limb weakness, and cyanosis; 21 h after admission, she had respiratory and cardiac arrest suddenly and died soon after; chest x-ray showed bronchopneumonia	Brainstem, spinal cord, hypothalamus, and glandular epithelium of intestines
3	2012	20/M	Wuhan Hubei	Fever, cough, and rhinorrhea for 2 d; he suddenly developed cyanosis, had a cardiac arrest, and died 16 h after admission; chest x-ray showed pulmonary edema	Brainstem, spinal cord, dentate nucleus, tonsil, glandular epithelium, and autonomic ganglia of intestines
4	2012	5/F	Huanggang Hubei	Lethargy and fever for 2 d associated with oral ulcers; she had pharyngeal hyperemia and lower-limb weakness; developed vomiting, cold extremities, and cyanosis and died 12 h after admission	Brainstem, spinal cord, dentate nucleus, hypothalamus, motor cortex, glandular epithelium, and autonomic ganglia of intestines
5	2012	30/F	Zhumadian Henan	Lethargy and fever for 4 d and intermittent seizures for 3 h associated with rash on four limbs; had several seizures and sudden respiratory arrest and died 23 h after admission	Brainstem, spinal cord, hypothalamus, and epithelium of intestines
6	2013	19/F	Yichang Hubei	Fever and vomiting for 3 d; developed cyanosis, tachypnea, and unconsciousness and died 5 h after admission	Brainstem, spinal cord, and epithelium of intestines

IHC, immunohistochemistry.

^aBrainstem included midbrain, pons, and medulla. Spinal cord tissues included attached peripheral nerve roots.

extracted from these specimens using the RNeasy Mini Kit (Qiagen, Hilden, Germany), followed by reversely transcribing into complementary DNA (cDNA) (One Step RNA PCR Kit; TaKaRa, Kyoto, Japan) according to the manufacturer's instructions. The procedure of nested RT-PCR consisted of two PCR cycles, with the first round of PCR using cDNA as a template and the second round of PCR using the amplicons from the first round of PCR as a template. The two round primers were designed from the whole VP1 sequence of the EV71 BrCr strain: in the first round, 5'-GGAGATAGGGTRGCRGATGTGATT-3' was the sense strand primer and 5'-RAGDGTRGTGATYGCYGTCTCGAC-3' was the antisense strand primer, and in the second round, 5'-YCCTCTWRAGGGCACMACYAA-3' was the sense strand primer and 5'-TCKGAYRRYTTGACAAAACTGAGG-3' was the antisense strand primer (key to degenerate nucleotides: M = A + C, R = A + G, W = A + T, Y = C + T, K = G + T, and D = A + G + T). RD cells (human rhabdomyosarcoma cells), provided by the Microorganisms & Viruses Culture Collection Center (Wuhan Institute of Virology, Chinese Academy of Sciences), were used for EV71 isolation as reported previously.²⁹

Results

Gross Findings and Pathologic Examination

All brains of the cases showed brain edema grossly. Histologic examination revealed severe edema and acute encephalomyelitis and scattered mild meningitis **Image 1A**. Inflammatory changes consisted of perivascular cuffs, microglia nodules **Image 1C**, parenchymal infiltration, degeneration, and necrosis with neuronophagia in some neurons **Image 1B**. In some areas, swelling astrocytes and microabscess-like formation were found. The most prominent inflammatory lesions were in the gray matter, including the entire medulla, pontine tegmentum, midbrain, thalamus, dentate nucleus, and anterior and posterior horns of the spinal cord. Multiple or focal encephalomalacia with foam cells in the brainstem and spinal cord was detected in five cases, two of which had symmetrical encephalomalacia **Image 1D**. Less inflammation was also seen in the cerebral motor cortex in three cases and basal ganglia in two cases. No viral inclusions were seen.

Varying amounts of lesions were seen in non-CNS tissues. Gross inspection and histologic examination both revealed marked pulmonary edema **Image 1E** in all cases. Multifocal

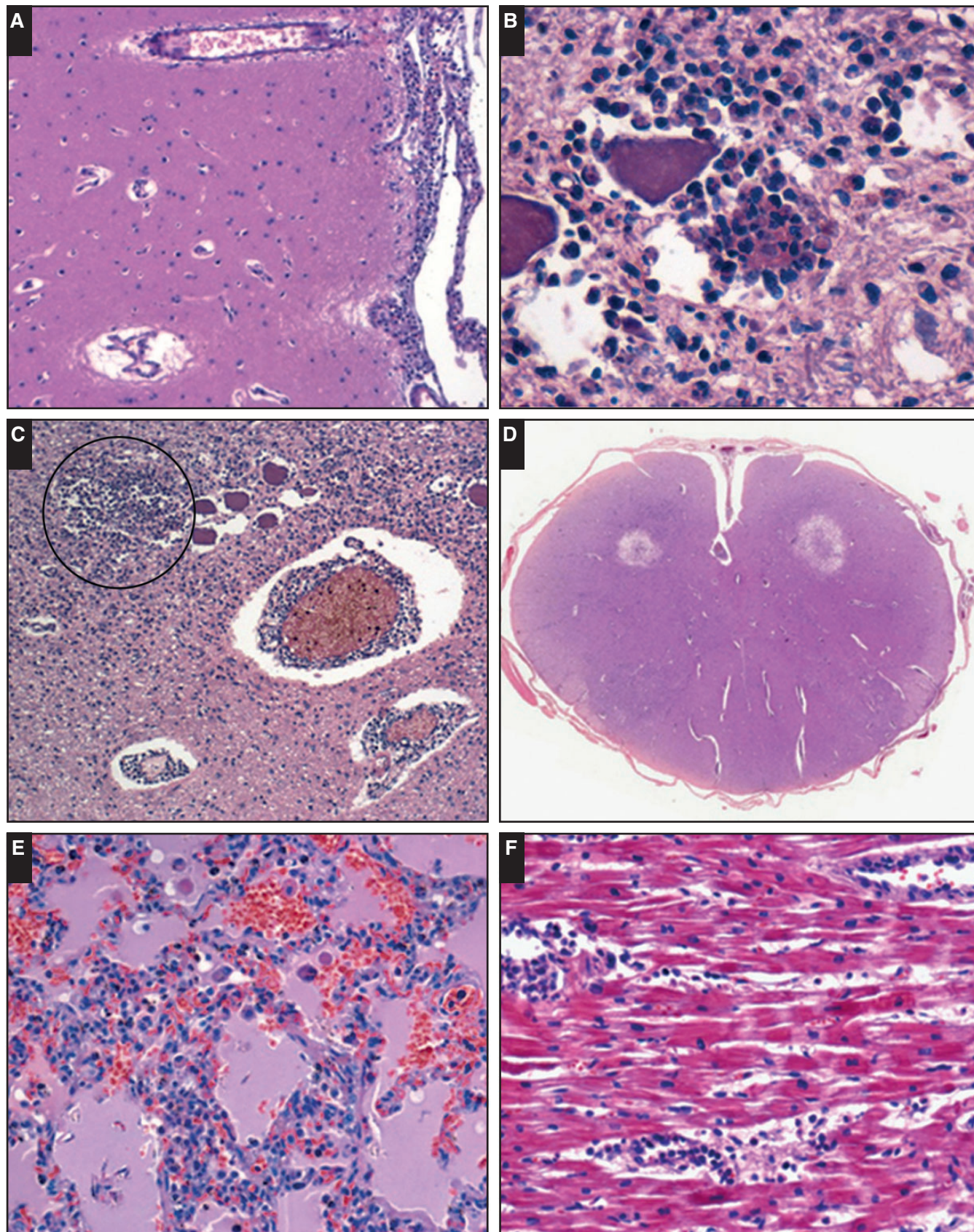


Image 1 Pathologic findings in fatal cases of enterovirus 71 infection. **A**, Mild meningitis and remarkable brain edema in the motor cortex. **B**, Neuronal necrosis and neuronophagia in the medulla oblongata. **C**, Perivascular cuffing and microglia nodule (circle) in the pons. **D**, Symmetrical encephalomalacia with foam cells in the cervical spinal cord. **E**, Remarkable pulmonary edema and focal pulmonary hemorrhage. **F**, Mild inflammatory cell infiltration associated with myocardial contraction band necrosis in the left ventricle of the heart.

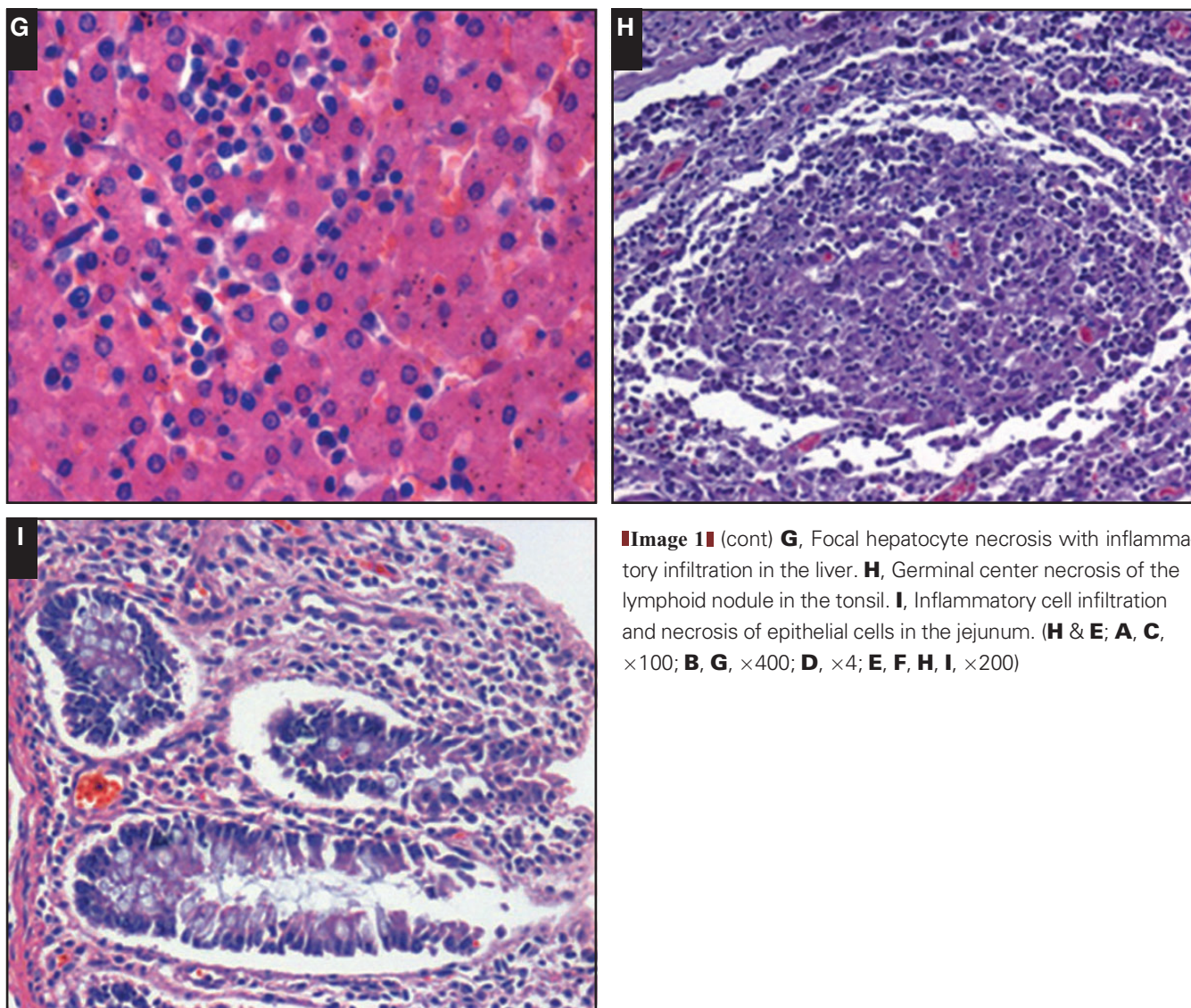


Image 1 (cont) **G**, Focal hepatocyte necrosis with inflammatory infiltration in the liver. **H**, Germinal center necrosis of the lymphoid nodule in the tonsil. **I**, Inflammatory cell infiltration and necrosis of epithelial cells in the jejunum. (**H & E**; **A, C**, $\times 100$; **B, G**, $\times 400$; **D**, $\times 4$; **E, F, H, I**, $\times 200$)

hemorrhage (Image 1E), hyaline membrane formation, and mild to moderate inflammatory infiltration were also seen. Three cases had mild myocarditis and endocarditis, manifesting as focal inflammatory infiltration and myocardial contraction band necrosis in the myocardium (Image 1F). Hepatocyte edema and microvesicular fatty change were seen in all cases, and focal hepatocyte necrosis was seen in one case (Image 1G). Enlarged tonsils and mesenteric lymph nodes as well as reactive lymphoid hyperplasia and necrosis (Image 1H) were seen in four cases. Varying degrees of enteritis (Image 1I), presented as inflammatory cell infiltration in the mucosa and submucosa, were seen in all cases. No inflammatory infiltration was observed in the CNS and non-CNS tissues of the negative control case.

Distribution of Viral Antigens

Viral antigens were expressed to varying degrees in the cytoplasm of neurons, neuronal processes, and microglia

nodules in different parts of the CNS, particularly in the areas containing severe damage and inflammation, such as the brainstem, dentate nucleus, thalamus, basal ganglia, and spinal cord, including posterior and anterior horn cells (Image 2A), (Image 2B), and (Image 2D). Detailed analysis revealed that the viral antigens were mainly located in the motor nuclei of the brainstem of all cases. The motor cortex, cingulate gyrus, and nerve roots of spinal cord also showed mild to moderate positive staining (Image 2C), (Image 2E), and (Image 2F). In the two cases with reactive lymphoid hyperplasia and necrosis, the expression of viral antigens in the CNS was much less and sporadic compared with other cases, although the inflammation was severe. Scattered inflammatory cells in the tonsils, macrophages in the submucosa and mesenteric lymph nodes of intestines, epithelial cells of Peyer's patches, and epithelial cells of intestinal glands in the duodenum, jejunum, ileum, and colon also demonstrated positive viral antigens (Image 2G) and

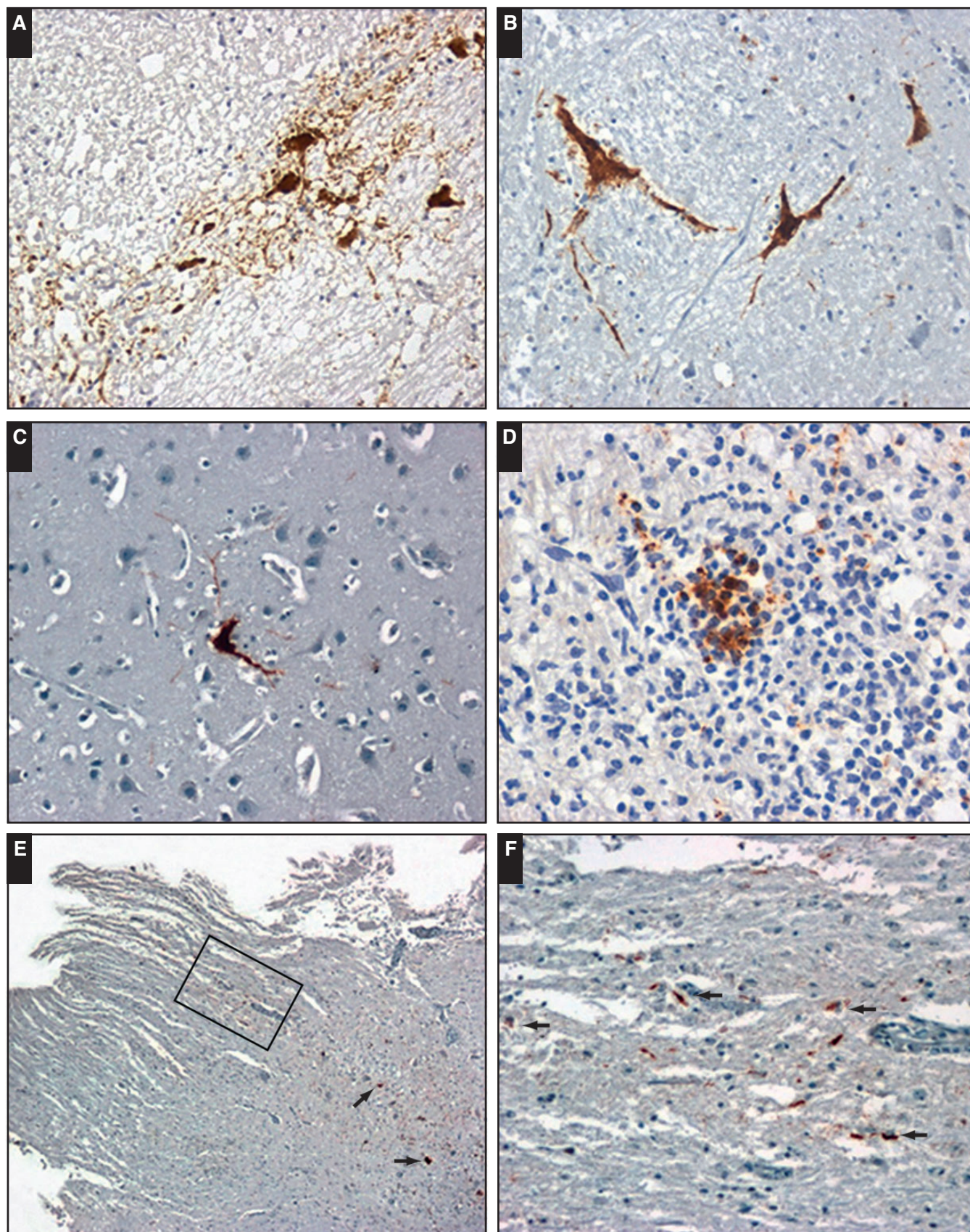


Image 2 Distribution of viral antigens in various tissues. The primary antibody used in immunohistochemistry (IHC) was a monoclonal antibody against the VP1 protein of enterovirus 71. **A**, In the neuronal bodies and processes in the pons. **B**, In the neurons of posterior spinal cord horn. **C**, In a single neuron in the motor cortex of the cerebrum. **D**, In a microglia nodule in the medulla. **E**, In the spinal cord neurons (arrows) and attached peripheral nerve root (rectangle). **F**, Higher amplification of the peripheral nerve root showed positive viral antigen staining (arrows).

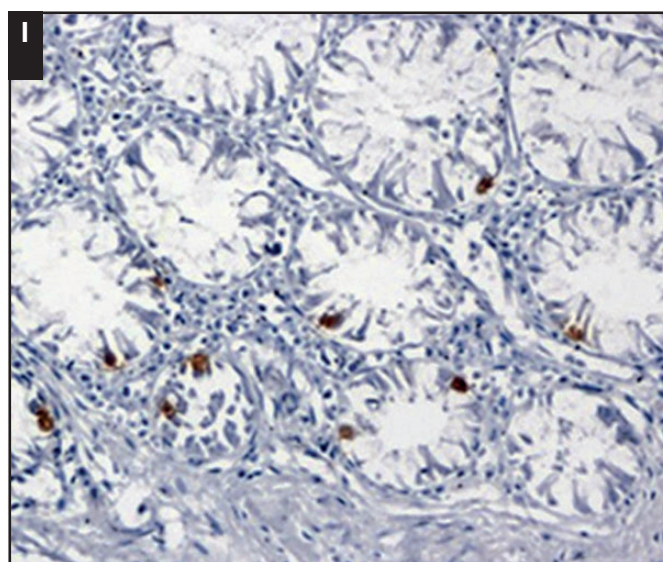
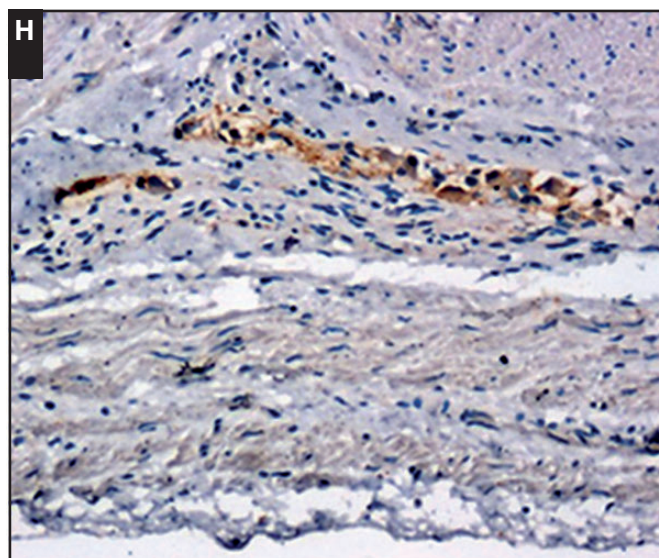
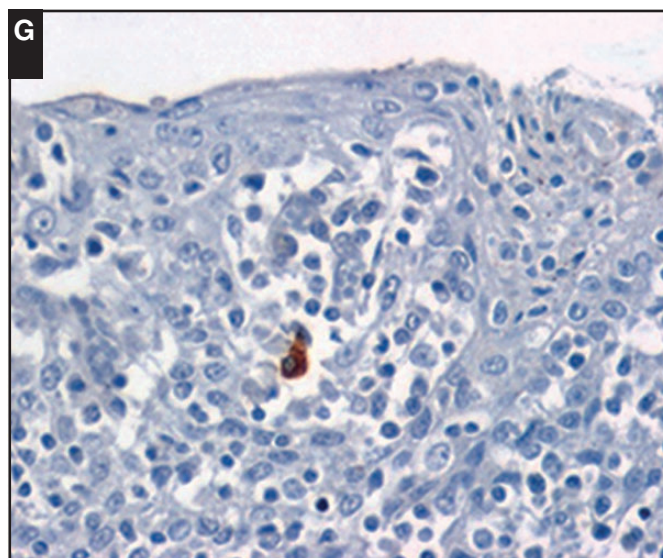


Image 2I (cont) **G**, In an inflammatory cell in the tonsil. **H**, In the autonomic ganglia of the small intestine. **I**, In the glandular epithelial cells of the colon. (IHC; **A**, **C**, **F-I**, $\times 100$; **B**, **D**, $\times 400$; **E**, $\times 40$)

Image 2II. In particular, the autonomic ganglia of intestines showed positive viral antigens **Image 2HI**. No viral antigens were found in the ependyma, choroid plexus, and blood vessels of the brain and other non-CNS tissues. In the negative control case, neither the CNS nor the non-CNS tissues expressed viral antigens.

Expression of Viral Receptor SCARB2

In both of the EV71-infected patients and negative control case, SCARB2 was widely expressed in the neurons, microglia cells, and some inflammatory cells in the cerebrum, cerebellum, brainstem, and spinal cord. In the non-CNS, SCARB2 was positive in macrophages in the immune organs and lung, trachea, epithelial cells of salivary glands, and glandular epithelial cells of the intestine. However, SCARB2 was weakly expressed or negative in skeletal muscle, cardiac muscle, liver, kidney, pancreas, and thyroid (data not shown).

Systemic Inflammatory Cell Infiltration

In the CNS, the most inflammatory cells were CD68+ macrophages/microglia and CD15+ neutrophils, which were distributed widely in the areas with severe inflammatory changes, especially in the microglia nodules and perivascular cuffs as previously mentioned in all cases **Image 3A** and **Image 3B**. The CD20+ B cells were mainly in the perivascular cuffing and meninges, although they were less concentrated **Image 3C**.

In the non-CNS, the predominant inflammatory cells were also CD68+ macrophages and CD15+ neutrophils, which were mainly located in the intestinal mucosa, lung, and heart with mild myocarditis **Image 3DI**, **Image 3EI**, and **Image 3FI**. CD68+ macrophages and CD15+ neutrophils were also mildly to moderately positive in the gastric mucosa and bronchial wall but negative in the other tissues. Sporadic CD20+ B cells were found in the gastrointestinal mucosa, bronchial wall, and alveolar wall but negative in

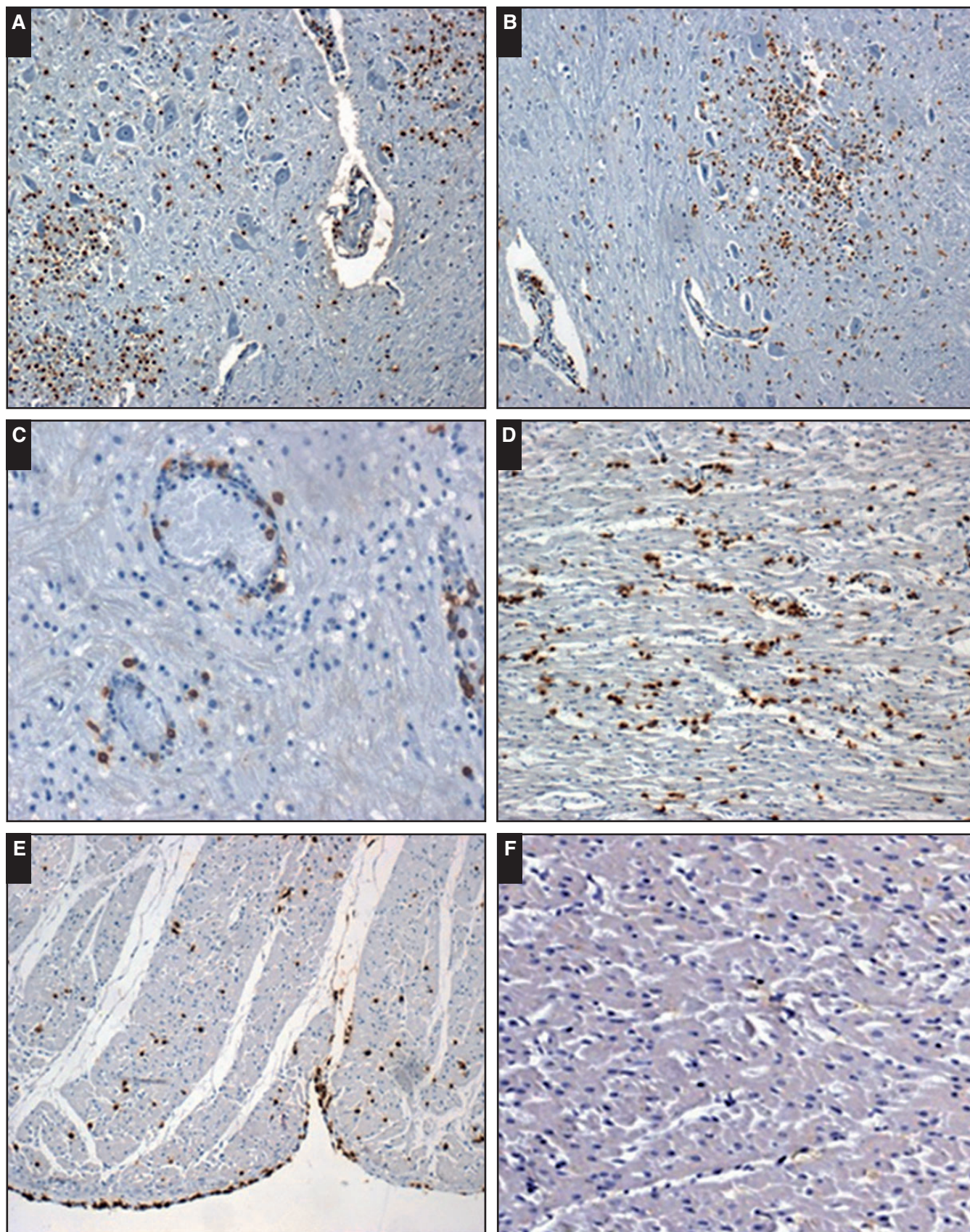


Image 3 Immunohistochemistry demonstrated CD68+ macrophages/microglia, CD15+ neutrophils, and CD20+ B cells in the brainstem (**A-C**) and heart (**D, E**). No CD20+ B cells were detected in the heart (**F**). (**A, B, E**, $\times 100$; **C, D, F**, $\times 200$)

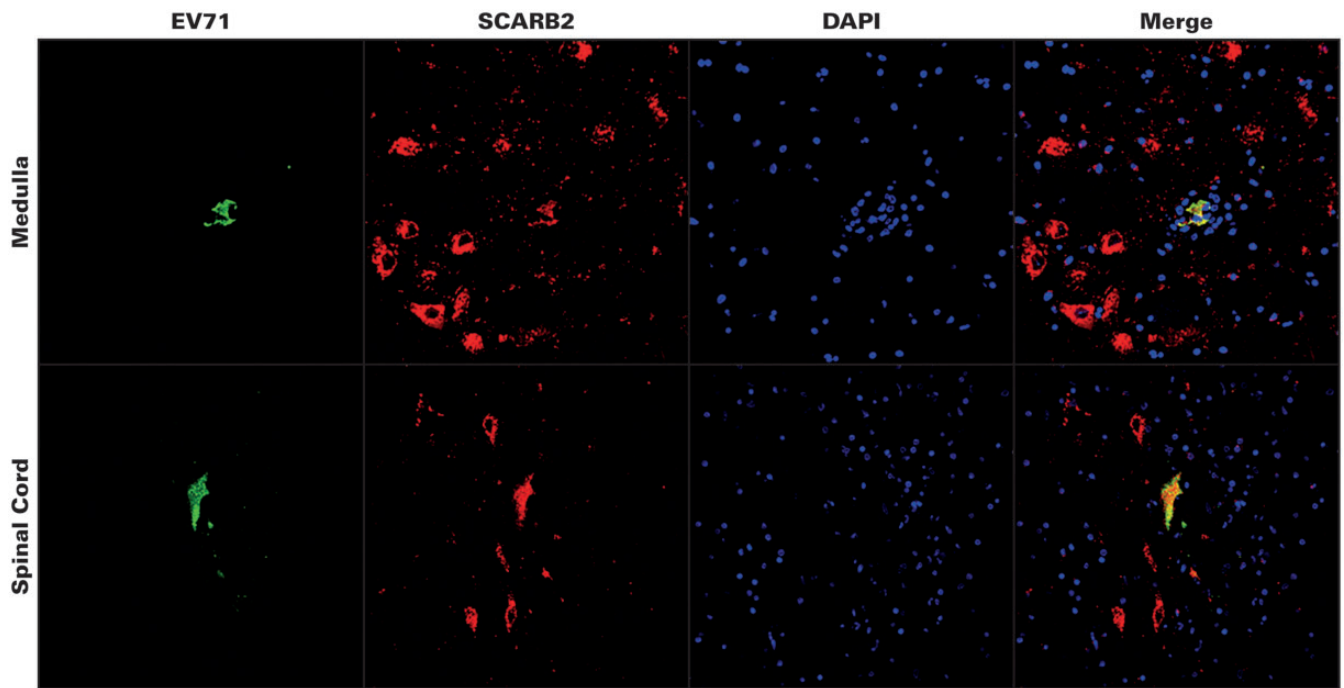


Image 4 Double immunofluorescence staining of viral antigens with SCARB2. A neuron with neuronophagia in the medulla and a neuron in the anterior spinal cord horn both showed the colocalization of viral antigens and SCARB2. ($\times 400$)

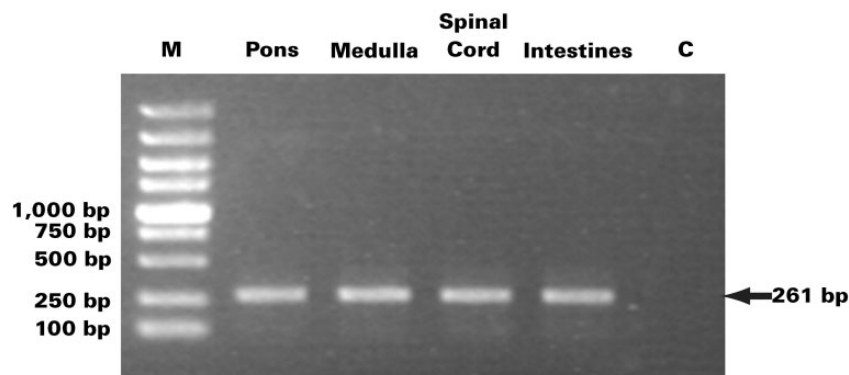


Image 5 Enterovirus-specific reverse transcription polymerase chain reaction products with a predicted size of 261 base pairs were identified in RNA extracts from the fresh pons, medulla, spinal cord, and intestines. M, DNA ladder; C, negative control.

the heart (Image 3F). The CD3⁺ T cells and CD56⁺ natural killer cells were either less or absent in the CNS and non-CNS. In the negative control case, no inflammatory infiltration was observed in the CNS and non-CNS except for the lymphoid tissues.

Double Immunofluorescent Staining for Viral Antigens With SCARB2

To confirm that EV71 infection was dependent on SCARB2 expression, we used viral antigen-positive tissues, including the brainstem and spinal cord, for

double immunofluorescent staining. The results showed that all viral antigen-positive cells expressed SCARB2

Image 4

Confirmation of EV71 Infection by Nested RT-PCR

Enterovirus-specific nested RT-PCR products with a predicted size of 261 base pairs **Image 5** were identified in RNA extracts from all cases. The sequencing results confirmed the EV71 infection (data not shown). The nested RT-PCR-positive specimens included brainstem, spinal cord, motor cortex, cerebellum, thalamus, basal ganglia,

cerebrospinal fluid, trachea, small intestine, large intestine, mesenteric lymph nodes, oral swab, throat swab, and anal swab (data not shown). In addition, we successfully isolated four EV71 strains from the fresh brainstem tissues of four cases (data not shown).

Discussion

The results of IHC, nested RT-PCR, and virus isolation confirmed that all six patients died of acute EV71 encephalomyelitis. The nested RT-PCR–positive specimens had a greater scope compared with the findings by IHC. The possible cause is that the nested RT-PCR had a higher sensitivity than the IHC. Similar to previous studies,^{12-16,19-21} our results demonstrated that EV71 is highly neuronotropic, because viral antigens were mainly located in the neuronal cytoplasm and processes in different parts of the CNS and the autonomic ganglia of the intestines. Positive viral antigens in the macrophages in the tonsils and intestines, epithelial cells of the intestinal glands, and Peyer's patches indicated a link from the non-CNS to the CNS.

Previous studies suggest that polioviruses can invade the CNS through a disrupted blood-brain barrier or retrograde axonal spread along cranial or peripheral nerves.¹ For EV71, studies implicate the latter route, likely the motor but not the peripheral sensory or autonomic pathways because the dorsal root and autonomic ganglia were not found to be involved.^{16,19} Our study showed that viral antigens were expressed mainly in the anterior spinal cord horn neurons (Image 2B), motor nuclei in the brainstem corresponding to cranial nerves with motor components (3rd to 12th nerves) (Image 2A), and occasionally the motor cortex (Image 2C). These results suggest that the motor pathway is the major pathway for EV71 entry into the CNS. In addition, viral antigens in the posterior spinal cord horn neurons, similar to previous findings,¹² and the autonomic ganglia of the intestines (Image 2H) indicated that in addition to the major motor pathway, EV71 can also enter the CNS via peripheral sensory and autonomic pathways in retrograde axonal transport. Positive viral antigens in the spinal nerve roots (Image 2E,F) and the autonomic ganglia (Image 2H) showed the direct neuropathologic evidence to support viral entry into the CNS through peripheral nerves.

Viral entry into susceptible host cells is dependent on specific receptors. SCARB2 is considered a critical receptor to all EV71 strains.³⁰ In our study, SCARB2 was observed to have a widespread distribution in various tissues, consistent with previous studies.^{15,28} Double immunofluorescence staining demonstrated that all viral antigen–positive cells in the brainstem and spinal cord expressed SCARB2 (Image 4). These results further confirmed SCARB2 as a functional receptor for EV71 infection and suggested that SCARB2

expression in the susceptible cells is the basis of EV71 infection. In addition, positive expression of SCARB2 and viral antigens in the large and small intestines supported the intestinal tract as being the primary site of EV71 infection. We also noticed that although many cells were SCARB2 positive, virus infection appeared to be located in specific foci (Image 4), indicating that EV71 is transmitted through a specific pathway instead of random spreading to the surrounding cells.

On the basis of these findings, we speculate the possible pathways for virus entry into the CNS: EV71 initially infects the intestinal tract mediated by the viral receptors, especially SCARB2, and then retroactively infects the nerve endings and autonomic ganglia, which control the digestive tract, and subsequently enters into the spinal cord and brainstem via the peripheral nervous system, including the motor, sensory, and autonomic pathways. Although the hematogenous route for viral spread into the CNS could not be excluded, our data did not support this pathway, because no viral antigens were found in the vascular endothelial cells, ependyma, and choroid plexus.

Previous studies have generated some controversies as to which inflammatory cells predominate in the CNS during EV71 infection. Some studies considered that the inflammatory cells were predominantly neutrophils,^{2,17} but others regarded macrophages/microglia as the predominant inflammatory cells,^{12,14,16} and neutrophils were sparse or even negative.¹² Our results revealed that although CD68+ macrophages/microglia were the predominant inflammatory cells, CD15+ neutrophils were also abundant. In contrast, the CD20+ B cells and CD3+ T cells were less. A clinical study revealed that patients with EV71 who had pulmonary edema had depleted levels of T lymphocytes in peripheral leukocytes and reduced cellular immunity,³¹ possibly contributing to the paucity of T cells in the CNS,¹⁶ as in this study. These results indicated that inflammatory cells play a significant role in the CNS infection.

The inflammation also involved the immune system, gastrointestinal tract, lung, and heart. Similar to the CNS, the predominant inflammatory cells in these tissues were CD68+ macrophages/microglia and CD15+ neutrophils. Although previous case reports mostly showed no histologic evidence of inflammation in cardiac tissue,^{2,3,5,16} mild myocarditis¹⁸ and focal inflammatory infiltration in the myocardium^{12,15,17} have been reported. In the two cases with extensive lymphatic tissue hyperplasia and necrosis, which indicated immune function impairment, although the encephalomyelitis was significant, viral antigens were much less than in other cases. Infants and young children are very susceptible to EV71 infection, and immature immunity is suspected to be associated with the high morbidity and mortality.³² Animal studies also indicated that the host immune

responses are related to the serious complications of EV71 infection.³³ The above results, including our findings, suggest that in addition to CNS inflammatory response, both the systemic inflammatory response and host immune response play important roles in the pathogenesis of EV71 infection. In particular, the inflammatory response and host immune function impairment may potentially play a more important role than the direct damage caused by EV71 infection in some cases. Additional experiments, such as identifying inflammatory factors in different tissues, should help to further investigate the direct or indirect damages caused by EV71 infection.

Pulmonary edema was speculated to be neurogenic, caused by lesions of the vasomotor center in the brainstem.^{1,3,12} Our results suggested that, in addition to a neurogenic origin, possible increased vascular permeability secondary to the systemic inflammatory response and raised pulmonary pressure caused by cardiac damage may both contribute to pulmonary edema.

In conclusion, our findings revealed direct pathologic evidence supporting viral entry into the CNS through peripheral nerves. In addition to the major motor pathway, EV71 can also possibly enter the CNS via peripheral sensory and autonomic pathways in retrograde axonal transport. Inflammatory cells in the CNS and non-CNS were both predominantly CD68+ macrophages/microglia and CD15+ neutrophils. The CNS and systemic inflammatory response, as well as host immune response, contributes to the pathogenesis of EV71 infection. The present findings will enhance our understanding of the pathogenesis of EV71.

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